ACADEMIC CENTRAL

PRACTICE EXAMINATION

**NC(V) LEVEL 2**

# ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

## PRACTICE EXAM

**MARKS: 100** **TIME: 3 hours**

**NC(V) Programming and Robotics, Level 2. Section A: objective questions (30) · Section B: safety and DC principles, Ohm's law and power, components, number systems, logic gates and truth tables (70).**

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training examination paper.

## INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Show ALL formulae, substitutions and calculations, and write units with all answers.
5. A non-programmable calculator may be used.
6. Write neatly and legibly.

## SECTION A

## QUESTION 1

**1.1** Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10).

1.1.1 The output of an AND gate is 1 only when …

A any input is 1. B all inputs are 0.
C the inputs are different. D all inputs are 1.

1.1.2 A NOT gate is also called an …

A amplifier. B inverter. C adder. D encoder.

1.1.3 The decimal number 13 in binary is …

A 1011. B 1110. C 0111. D 1101.

1.1.4 Which component is used to detect light in a line-following robot?

A Capacitor B LDR or phototransistor
C Fuse D Inductor

1.1.5 A robot's 9 V battery supplies 0,3 A. The power used is …

A 30 W. B 0,03 W. C 2,7 W. D 9,3 W.

1.1.6 In digital electronics, a logic 1 usually represents …

A a high voltage, e.g. 5 V. B 0 V.
C a negative voltage. D an open circuit only.

1.1.7 The number of possible input combinations for a gate with three inputs is …

A 3. B 8. C 6. D 9.

1.1.8 A device that converts electrical energy into rotary motion in a robot is a …

A sensor. B switch. C resistor. D motor.

1.1.9 The output of an OR gate with inputs A = 0 and B = 1 is …

A 1. B 0. C 2. D undefined.

1.1.10 A component that stores energy in an electric field is a …

A diode. B transistor. C capacitor. D relay.

(10)

**1.2** Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'true' or 'false' next to the question number (1.2.1–1.2.10).

1.2.1 A NAND gate gives the opposite output of an AND gate.

1.2.2 Binary uses the digits 0 to 9.

1.2.3 Resistors in series share the same current.

1.2.4 A switch is an example of a digital input.

1.2.5 An LED will light up regardless of which way it is connected.

1.2.6 A transistor can be used as an electronic switch to control a motor.

1.2.7 The total resistance of two 10 Ω resistors in parallel is 20 Ω.

1.2.8 A truth table shows the output for every possible combination of inputs.

1.2.9 A multimeter on the voltage range is connected in parallel with the component.

1.2.10 One byte is made up of 4 bits.

(10)

**1.3** Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (1.3.1–1.3.10).

| | **COLUMN A** | | **COLUMN B** |
|---|---|---|---|
| 1.3.1 | Output is 1 when the inputs are different | A | soldering |
| 1.3.2 | Output is 1 only when all inputs are 0 | B | hexadecimal |
| 1.3.3 | Signal with only two states, on and off | C | ultrasonic sensor |
| 1.3.4 | Signal that can have any value within a range | D | analogue signal |
| 1.3.5 | Electrically operated switch that uses an electromagnet | E | NOR gate |
| 1.3.6 | Measures distance by sending out sound waves | F | XOR gate |
| 1.3.7 | Board with holes used to build test circuits without soldering | G | relay |
| 1.3.8 | Variable resistor turned by a knob | H | potentiometer |
| 1.3.9 | Number system with base 16 | I | breadboard |
| 1.3.10 | Joining components with melted metal alloy | J | digital signal |

(10)

**TOTAL SECTION A: [30]**

## SECTION B

## QUESTION 2: SAFETY AND DC PRINCIPLES

**2.1** Give FOUR safety rules for soldering. (4)

**2.2** Explain what current, voltage and resistance are, and give the unit of each. (6)

**2.3** State Ohm's law and write down the formula. (2)

**2.4** A robot's LED is connected to a 5 V supply through a resistor. The LED needs 2 V and 20 mA. Calculate the value of the series resistor. (4)

**2.5** A DC motor draws 1,5 A from a 12 V battery. Calculate: (a) the power; (b) the motor's resistance. (4)

**[20]**

## QUESTION 3: COMPONENTS AND CIRCUITS

**3.1** Three resistors of 100 Ω, 220 Ω and 330 Ω are connected in series across a 9 V supply. Calculate: (a) the total resistance; (b) the current; (c) the voltage across the 220 Ω resistor. (6)

**3.2** Two resistors of 60 Ω and 30 Ω are connected in parallel. Calculate the total resistance. (3)

**3.3** Draw the circuit symbols of: (a) an NPN transistor; (b) a DC motor; (c) a push-button switch (normally open). (3)

**3.4** Explain why a diode is connected across (in parallel with) a motor or relay coil that is switched by a transistor. (3)

**[15]**

## QUESTION 4: NUMBER SYSTEMS

**4.1** Convert: (a) decimal 45 to binary; (b) binary 1110 0101 to decimal; (c) binary 1010 1111 to hexadecimal. (6)

**4.2** Add the binary numbers 0110 1011 and 0001 1101. (2)

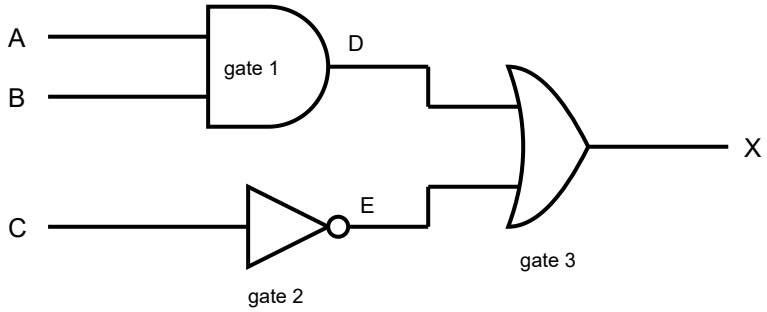
**4.3** A robot uses an 8-bit sensor value. What is the largest decimal value it can store? Show how you got the answer. (2)

**[10]**

## QUESTION 5: LOGIC GATES

**5.1** Draw the symbol and the truth table for a two-input NAND gate. (5)

**5.2** Study the logic circuit below. A robot must stop (X = 1) when both bump sensors A and B are pressed, or when the safety key C is removed (C = 0).



5.2.1 Name gates 1, 2 and 3. (3)

5.2.2 Write the Boolean expression for X. (2)

5.2.3 Complete the truth table for A, B, C, D, E and X (eight rows). (8)

5.2.4 Use your truth table: does the robot stop when A = 1, B = 0 and C = 1? Explain. (2)

**5.3** A robot arm may only move (Y = 1) when the guard door is closed (G = 1) AND either start button S1 or start button S2 is pressed. Write the Boolean expression for Y and draw the logic circuit. (5)

**[25]**

**TOTAL SECTION B: [70]**

**TOTAL: 100**

**END OF QUESTION PAPER**